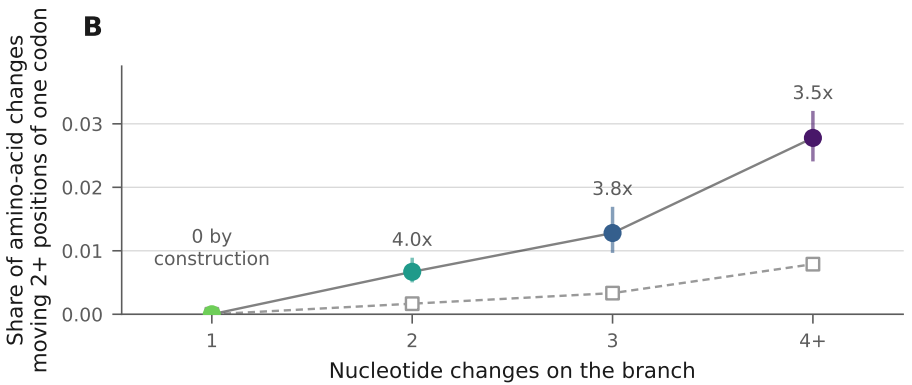
A

	Total	Fewest nucleotide changes needed 1	2	3
Individual internal branches	30,028	29,875 (99.5%)	147 (0.5%)	6 (<0.1%)
KY348531 to newest internal nodes	3,563,404	2,930,886 (82.2%)	537,399 (15.1%)	95,119 (2.7%)
Null, genetic code (J.2; 64 x 20, codon to aa)	10,754	3,334 (31.0%)	5,486 (51.0%)	1,934 (18.0%)
PLM null, Base model (ESM-2) (J.2, summed probability)	566	364.3 (64.4%)	174.6 (30.8%)	27.1 (4.8%)
PLM null, HA80 fine-tune (J.2, summed probability)	566	496.3 (87.7%)	66.4 (11.7%)	3.3 (0.6%)

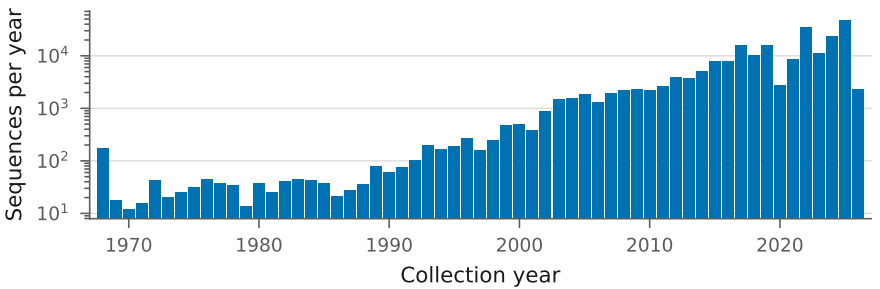
B



Observed
Expected by chance

Changes	branches	aa changes
1	33,648	12,813
2	9,311	6,878
3	3,508	3,750
4+	3,627	6,587

C



Tips	224,731
Years	1968-2026
Median year	2022
Busiest year	2025 (47,461)
From 2015	84%
Undated	0
Branches scored	50,094